



SYSTEM DESIGN

Detailed Step-by-Step Architecture

A layered view of the system, stage by stage — the clean data flow from raw stereo recordings to the manager's dashboard, and the naming used across the codebase.

1 • Data Ingestion Layer

Input: raw call data.

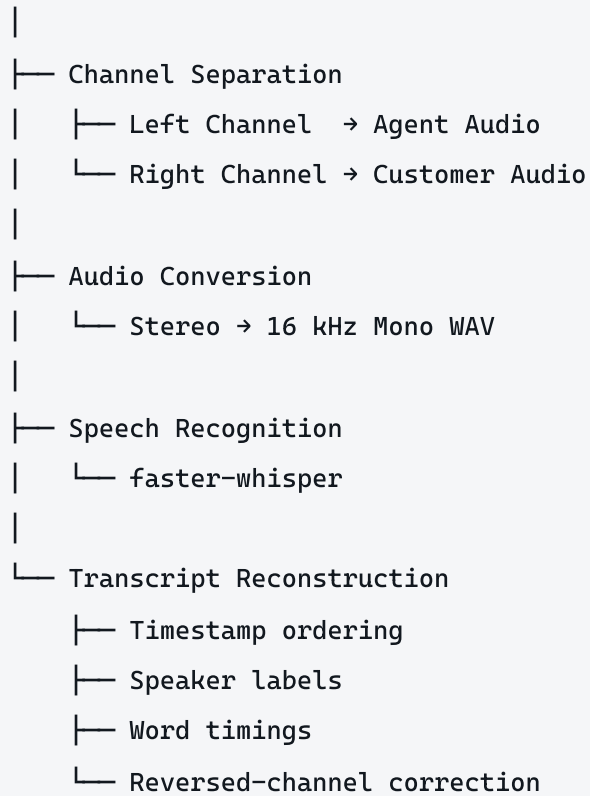
RAW CALL DATA

```
|
├─ Audio
|   └─ audio/<call_id>.mp3
|       └─ 8 kHz
|       └─ Stereo
|       └─ Left → Agent
|       └─ Right → Customer
└─ Metadata
    └─ metadata/<call_id>.json
        └─ Customer Name
        └─ Agent Name
        └─ Timestamps
```

2 • Speech Processing Layer

Speech-to-text & speaker separation.

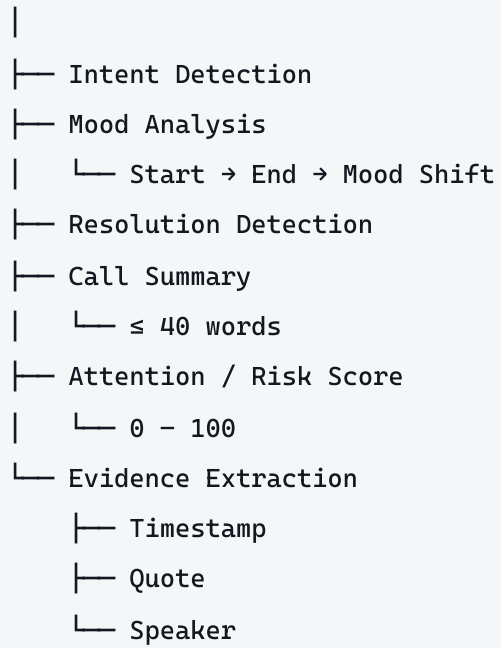
SPEECH PROCESSING



3 • Conversation Intelligence Layer

The analysis engine.

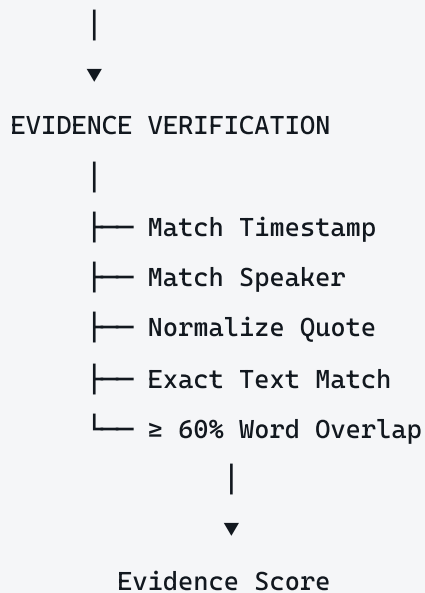
CONVERSATION INTELLIGENCE



4 • Evidence Validation Layer

Evidence verification engine.

AI ANALYSIS



Every AI judgment is checked against the actual transcript to verify that the supporting evidence is grounded in the conversation.

5 • Quality & Compliance Layer

Quality & compliance engine.

QUALITY & COMPLIANCE

|

├─ Greeting Check

├─ Identity Verification

├─ Action Confirmation

├─ Empathy Check

├─ Repeated Question Check

└─ Proper Closing

|

▼

QA Score (0–100)

|

▼

Resolution Risk

6 • Persistence Layer

Call intelligence database.

```
POSTGRESQL
|
└─ calls
    ├── Call Metadata
    ├── Transcript
    ├── Analysis
    ├── Evidence
    ├── Attention Score
    └─ QA Results
```

7 • Backend API Layer

CallRadar API.

```
CALLRADAR API
|
├─ /api/calls
├─ /api/qa
├─ /api/attention
└─ /api/eval
```

Data flow

```
PostgreSQL
  | READ ONLY
  ▼
FastAPI
  | REST API
  ▼
React + Vite
```

The API reads precomputed results from PostgreSQL. Calls are **not** re-transcribed during API requests.

8 • Frontend / Dashboard Layer

CallRadar Command Center.

CALLRADAR COMMAND CENTER

|

└─ Command Center

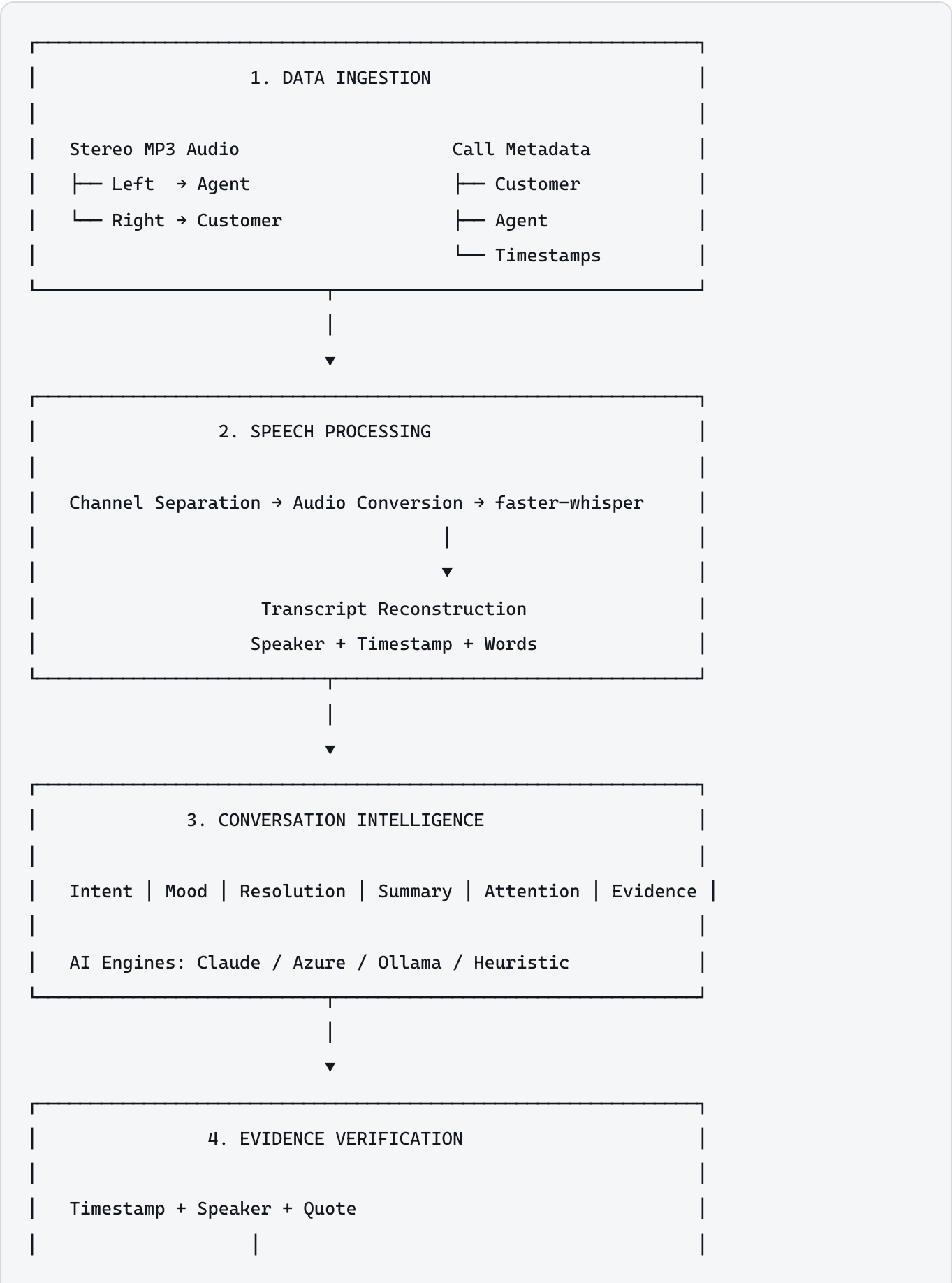
└─ QA Dashboard

└─ Customer View

└─ Call Details

└─ Quality / Evaluation

Final clean architecture



▼
Transcript Validation

|
▼
Evidence Score

5. QUALITY & COMPLIANCE ENGINE

Greeting | Identity | Confirmation | Empathy | Closing

|
▼
QA Score + Risk

6. CALL INTELLIGENCE DB

PostgreSQL

Transcript + Analysis + Evidence + QA

|
READ ONLY
|

7. CALLRADAR API

FastAPI

Calls | QA | Attention | Evaluation

8. CALLRADAR COMMAND CENTER

	React + Vite				
	Dashboard	QA	Customers	Call View	Quality

One-line pipeline

```
MP3 (Stereo)
  ↓
Channel Separation
  ↓
Whisper Transcription
  ↓
Transcript Reconstruction
  ↓
Conversation Intelligence
  ↓
Evidence Verification
  ↓
Quality & Compliance
  ↓
PostgreSQL
  ↓
FastAPI
  ↓
React Dashboard
```

Architecture naming

COMPONENT	RECOMMENDED NAME
Raw Data	Data Ingestion
Transcribe	Speech Processing
Analyze	Conversation Intelligence
Verify Evidence	Evidence Verification
QA / Compliance	Quality & Compliance
PostgreSQL	Call Intelligence Database
FastAPI	CallRadar API
React + Vite	CallRadar Command Center
Evaluate	Evaluation Engine

Companion for CallRadar — Detailed Step-by-Step Architecture. See also Architecture & design, Formulas, the complete documentation. | Team: Sahil Koshriya · Sakshi.